Mix criteria: complete manuscript costs

The historical article is Cunningham and Hubard, *Polytopality criteria for the mix of polytopes and maniplexes*, arXiv:2506.19334v1, all 21 physical pages. The comparison concerns its accepted corrected principal contract: 35 principal and background rows, with all three final problems still unanswered. One original is compared with one complete application in two papers, after paying for its entire common theory.

Whole topic. The primary ruler is literal untrimmed PDF extraction, NFC characters without whitespace, fixed before counting. Titles, abstracts, body, references, page footers and the source arXiv banner remain counted.

Complete document	Pages	Characters
Historical F005 source	21	57753
Common theory U	5	12615
Local application A_1	8	20788
Exceptional application A_2	4	10289
Both complete code appendices	$4 + 3$	16025

The written comparison charges both complete application papers:

$$R_{\text{written}} = \frac{57753}{12615 + 20788 + 10289} = \frac{57753}{43692} = 1.321821.$$

After acquisition of the *whole* common paper and *all three* complete mathematical programs, the one conditional comparison is

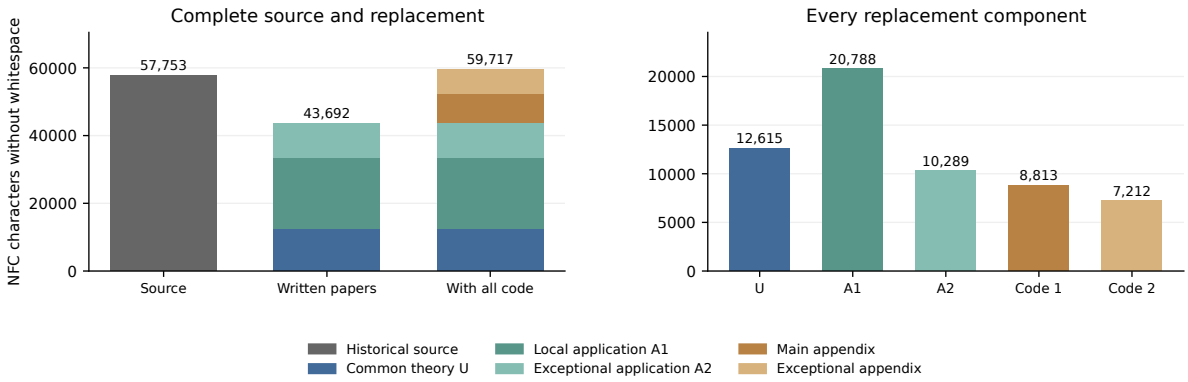
$$r \mid (U, P) = \frac{57753}{20788 + 10289} = \frac{57753}{31077} = 1.858384.$$

This conditional measure does not remove the acquisition from the whole-topic cost. The exceptional correction receives no free prior-work allowance; there is no overlap or cross-topic discount.

Whole program alternatives. One GAP file is a required finite certificate for the coprime-label example. Two full Python files validate written coordinate proofs. All three remain charged in these alternatives:

$$R_{\text{printed}} = \frac{57753}{43692 + 16025} = 0.967112, \qquad R_{\text{raw}} = \frac{57753}{43692 + 14608} = 0.990617.$$

Printed and raw representations are alternatives, never simultaneous charges. Both program-inclusive forms are longer than the source in this primary unit; the sensitivity table preserves the different results in other units. Three complete drivers cost 13674 characters separately; one full Lean file costs 1714 separately and contains two generic declarations. Neither is silently included in the displayed factors, and the full source is not formalized. Historical software cost is unknown, rather than zero.



Both axes start at zero. The rectangles show the actual complete costs in the primary unit, including both code appendices in the last total. The written replacement is shorter by 14061 characters in this representation. A single source supplies no recurring sawtooth pattern or theorem-node compression statistic.

Units and sensitivities. For complete untrimmed text in NFC:

Unit	Written	Conditional	Printed	Raw code
Whitespace tokens	1.442	2.026	1.232	1.258
Alphanumeric units	1.468	2.064	1.060	1.080
Nonspace characters	1.322	1.858	0.967	0.991
Characters, spaces collapsed	1.344	1.889	1.009	1.034
UTF-8 bytes, spaces collapsed	1.256	1.775	0.967	0.988

The JSON and CSV retain all five units under NFC and NFKC for untrimmed raw, selected raw and calibrated selected text: thirty coordinates and 120 exact fractions. The two selected sensitivities remove only isolated page footers; nine visually checked glyph aliases affect only the calibrated sensitivity. All source and replacement documents obey the same declared boundary.

Scientific and measurement limits. The accepted proofs include five literal counterexample targets, including Corollaries 6.10(3) and 6.15(2), and two further targets under the explicitly stated medial-component convention. Their corrected rooted criteria and retained classical inputs remain visible in the scientific source map. No answer to the three final problems, new source credit or historical originality claim is made by this measurement.

The accepted input audit binds 45 complete PDF pages, all three whole mathematical programs, and the separate driver and formal inventories. Extractable diagram labels and literal proxies are counted; full graphical information, external proofs, software runtime, engineering and full formalization costs remain unknown. These factors describe the declared text representation, not minimum proof complexity or total research effort. Distinct numerical review is required before publication.